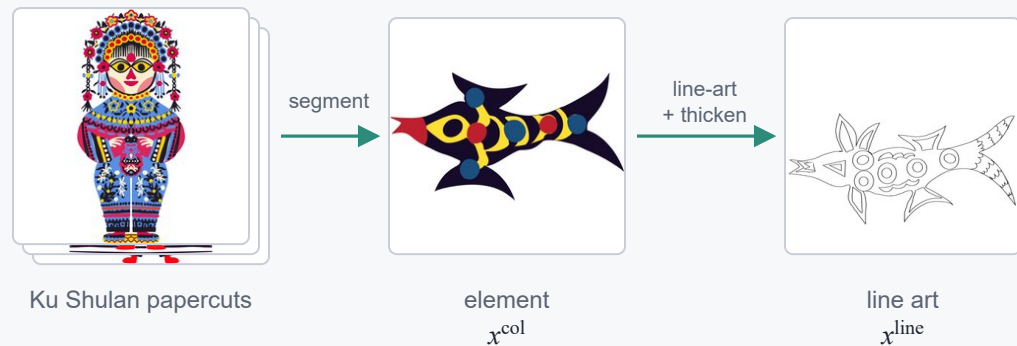


(a) Semi-structured Data Curation



Structured caption $y = (y_c, y_s)$

“The kushulan animal papercut is composed of geometric figures.

animal, fish, round_shapes, arc_shapes, simple_design, flat_design, red_background, green_accents, yellow_accents, red_yellow_combinations, geometric_patterns, sharp_lineart”

content tokens

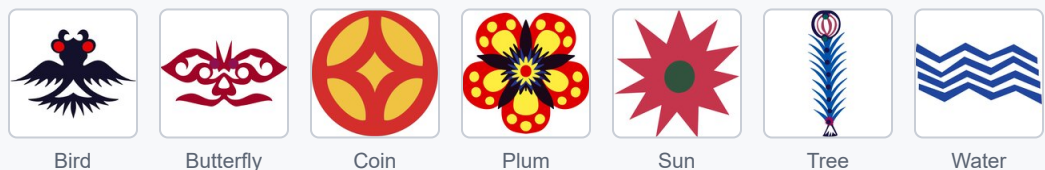
style / symbol tokens

source: DOCX prompt records

Six element categories



Pattern-symbol vocabulary

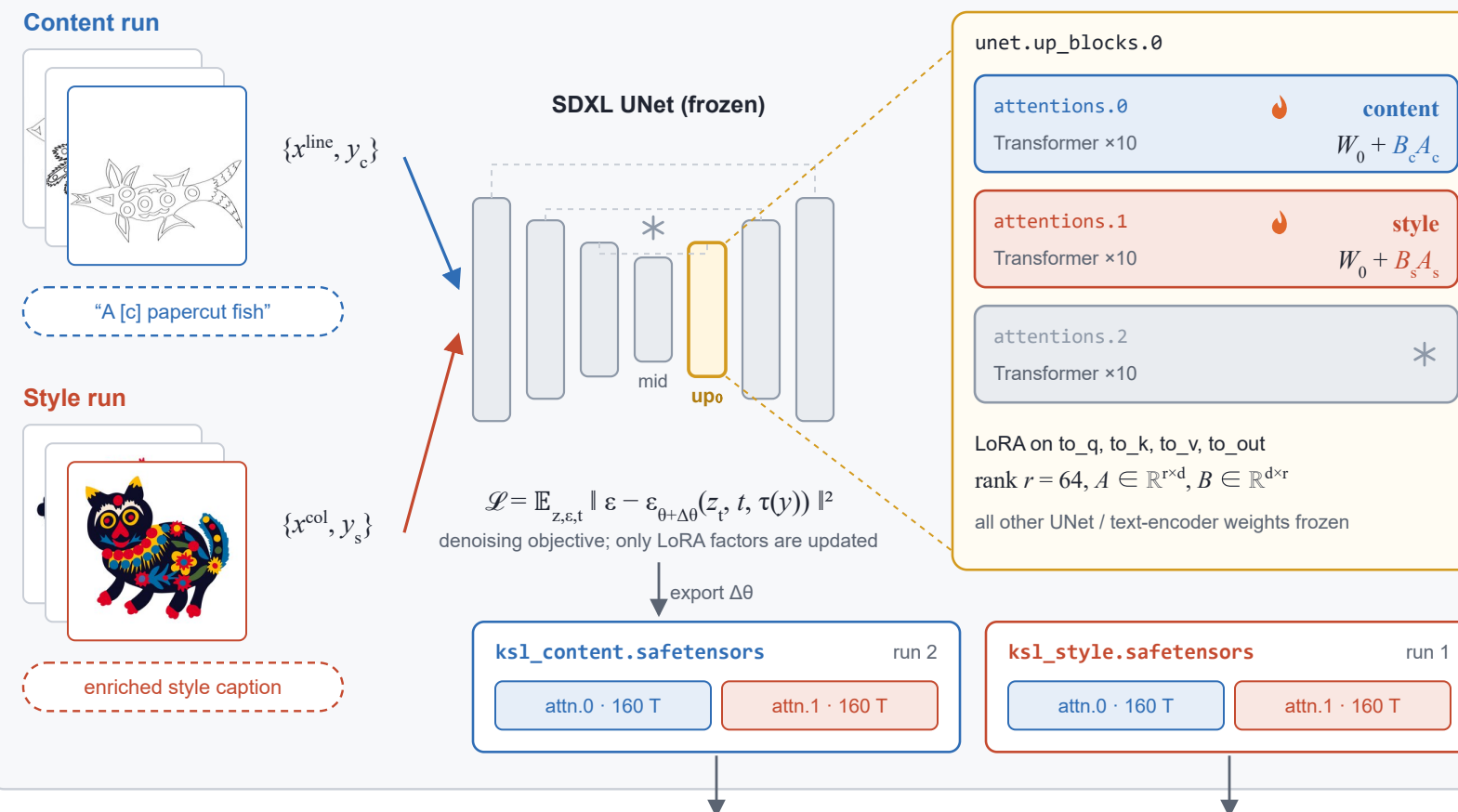


Record $r_i = \langle x^{\text{line}}, x^{\text{col}}, c, y, g \rangle$

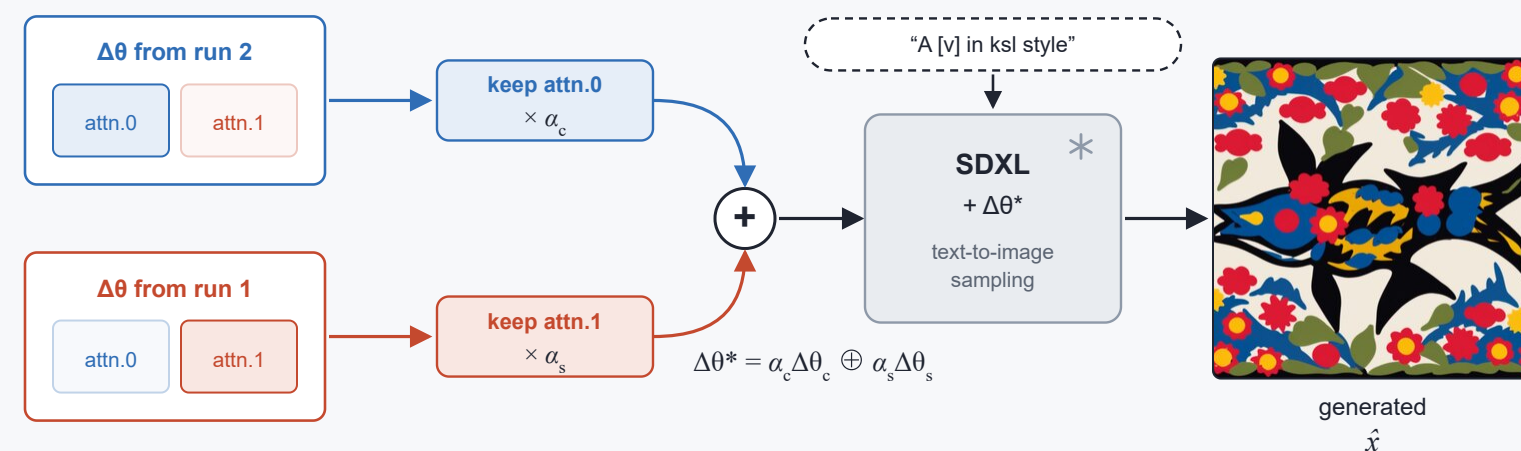
c: category / symbol class g: source-artwork group
282 curated images (179 elements + 103 symbols);
splits grouped by g to avoid near-duplicate leakage

* frozen 🔥 trainable (LoRA) □ content / structure □ style / appearance

(b) Block-wise B-LoRA Fine-tuning on SDXL

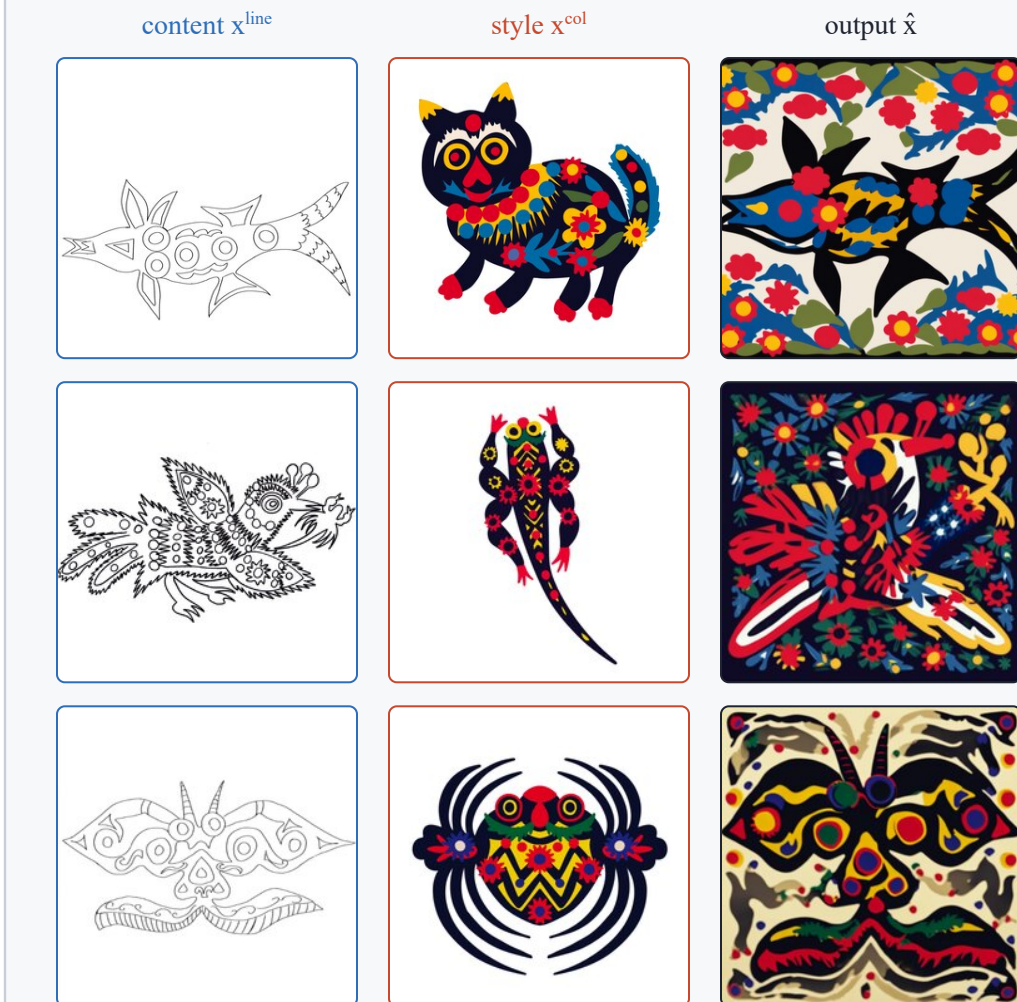


(c) Style–Content Composition at Inference



structure is inherited from the content adapter, colour / ornament from the style adapter; the two blocks are swapped independently across runs.

(d) Evaluation Protocol



archived qualitative results (7.5 run); not cherry-picked for this figure

Structure	$\text{DINO}(\hat{x}, x^{\text{line}}) \cdot \text{Edge-F1} / \text{Sil. IoU} \uparrow$
Style	$\text{CLIP-I}(\hat{x}, x^{\text{col}}) \cdot \text{colour-palette dist.}$
Text	$\text{CLIP-T}(\hat{x}, y)$
Human	structure · element · style · symbol (1–5), Kendall’s W

† pixel-aligned; reported with a mismatched-content control and counted only when matched > control (95% bootstrap CI)

tools/generate.py (fixed seeds) → tools/evaluate.py (per-sample CSV)